Micromachining Copper and Brass

In this article, we'll take a closer look at micromachining copper and brass, delving into their benefits, applications, and the techniques involved.

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Micromachining has emerged as a widely sought after manufacturing process for creating intricate parts on a microscopic scale. Among the various materials used in micromachining, copper and brass have grown in popularity for their exceptional properties, including for example: high thermal and electrical conductivity, corrosion resistance, and antimicrobial characteristics. In this article, we'll take a closer look at micromachining copper and brass, exploring their benefits, applications, and the techniques involved to produce micro manufactured parts.

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Benefits of Micromachining Copper and Brass

Excellent Conductivity

Copper exhibits exceptional electrical and thermal conductivity. Thus, micromachining copper allows for the creation of complex and intricate electrical components, such as microelectrodes, and connectors. Its superior conductivity ensures efficient signal transmission and reliable performance.

Corrosion Resistance

Copper possesses inherent corrosion resistance, and hence is suitable for applications exposed to various environments and corrosive substances. Micromachined copper components are commonly used in micro medical applications where corrosion resistance is vital for performance.

Antimicrobial Properties

Copper and brass have natural antimicrobial properties, known as the “antimicrobial effect.” This unique characteristic inhibits the growth of bacteria, fungi, and other harmful microorganisms on copper surfaces. Micromachined copper components find application in medical devices, such as antimicrobial catheters, wound dressings, and implantable devices, aiding in the prevention of infections.

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What are Common Grades of Copper and Brass for Micromachining?

Oxygen-Free Copper (C10100)

OFC exhibits high electrical conductivity and low impurity content, making it suitable for applications that require superior conductivity, such as microelectrodes and electrical connectors.

Beryllium Copper (C17200)

Beryllium copper alloys combine copper with a small amount of beryllium, resulting in improved strength, durability, and thermal conductivity. This grade is commonly used in micromachined components subjected to high stress, such as springs and electrical contacts.

Free Cutting Brass (C36000)

This widely used brass grade offers excellent machinability, making it suitable for intricate micromachined components. It furthermore exhibits good corrosion resistance and is often utilized in electrical connectors, fittings, and small precision parts.

Dezincification-Resistant Brass (C46400)

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This brass grade resists dezincification, a form of corrosion that occurs in some environments. It exhibits high corrosion resistance and is commonly used in plumbing and marine applications, as well as in micromachined components for medical devices.

Micromachining Techniques for Copper and Brass

Laser Micromachining

Laser micromachining involves the use of laser beams to precisely remove material from copper substrates, creating intricate features. This non-contact process thus offers high precision, minimal heat-affected zones, and excellent repeatability.

Electrical Discharge Machining (EDM)

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Micro EDM utilizes controlled electrical discharges to erode the copper material and form desired shapes. It is particularly suitable for complex geometries and hard-to-machine materials like copper.

Micro-Milling

Micro-milling employs small-diameter cutting tools to remove material and create precise features. Micro milling and turning commonly produces intricate features such as microcavities, channels, and intricate patterns on copper surfaces.

In need of micromachined Copper or Brass?

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